

CBSE Test Paper 05
Chapter 10 Mechanical Properties of Fluids

1. If a film of width L is stretched in the longitudinal direction a distance d by force F , surface tension is given by **1**
 - a. $S = F/4L$
 - b. $S = F/2L$
 - c. $S = F/3L$
 - d. $S = F/L$
2. In old age, arteries carrying blood in the human body, become narrow resulting in an increase in the blood pressure. This follows from **1**
 - a. Pascal's law
 - b. Stoke's law
 - c. Archimedes principle
 - d. Bernoulli's principle
3. If a pipe carrying incompressible liquid has an area A_1 and velocity V_1 at one point A_2 and V_2 at another point, the equation of continuity gives the relation **1**
 - a. $A_1 = A_2 V_2$
 - b. $A_1 V_1 = A_2 V_2$
 - c. $A_1 V_1 = A_2 V_2$
 - d. $A_1 V_1 = 2A_2 V_2$
4. A boy is carrying a bucket of water in one hand with a 0.05 kg cork piece in it. After sometime he takes out the cork piece and holds it in the other hand. The boy will carry **1**
 - a. Same load as before
 - b. Less load than before
 - c. More load than before
 - d. Either less or more load depending on the density of plastic
5. When the adhesive force in the case of liquid and glass is greater than the cohesive forces between the liquid molecules, the shape of the meniscus of liquid in a capillary tube is? **1**
 - a. Plane

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- b. Circular
 - c. Convex
 - d. Concave

6. The dams of the water reservoir are made thick near the bottom. Why? 1
 7. What is the value of bulk modulus for an incompressible liquid? 1
 8. Define surface tension? 1
 9. Briefly, explain, what do you mean by pressure exerted by a fluid. 2
 10. Why is a soap solution a better cleansing agent than ordinary water? 2
 11. The velocity of water in a river is 18 kmh^{-1} near the surface. If the river is 5 m deep, find the shearing stress between horizontal layers of water. The coefficient of viscosity of water 10^{-2} poise. 2
 12. Explain why? 3
 - i. The blood pressure in humans is greater at the feet than the brain.
 - ii. Atmospheric pressure at a height of about 6 km decreases to nearly half its value at the sea level though the height of the atmosphere is more than 100 km.
 - iii. Hydrostatic pressure is a scalar quantity even though the pressure is force divided by area.
 13. A horizontal pipe of diameter 20 cm has a constriction of diameter 4 cm. The velocity of water in the pipe is 2m/s and pressure is 10 N/m^2 . Calculate the velocity and pressure at the constriction? 3
 14. A vertical off-shore structure is built to withstand a maximum stress of 109 Pa. Is the structure suitable for putting up on top of an oil well in the ocean? Take the depth of the ocean to be roughly 3 km, and ignore ocean currents. 3
 15. What is meant by the term **coefficient of viscosity**. State Stoke's law. Define terminal velocity and find an expression for the terminal velocity in case of a small sphere falling through a viscous liquid such as glycerine. 5
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